STATISTICS · ANTICIPATING PATTERNS

Sampling Distribution Lab

Explore & Reflect

Name _____ Date _____ Period _____

Open **Sampling Distribution Lab** in your unit's Interactive Math section on Mr. K's Classroom (or use your teacher's link). Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3. Do Parts 4 and 5 on your own.

1 Predict before you touch the tool

1. A population of wait times is strongly right-skewed with $\mu = 6.85$ minutes and $\sigma = 5.66$ minutes. You take many samples of size 50 and record $\bar{x}$ each time. Predict the shape, center and spread of the $\bar{x}$ values.

2. What will the distribution of $\bar{x}$ look like if every sample has size $n = 1$?

2 Explore use the tool

1. Skewed population, $n = 1$: draw 1,000 samples. Describe the shape, center and spread of the $\bar{x}$ values.

2. Increase n until the skewness of the sample means is below 0.35 (1,000+ samples). Record n and describe the shape.

3. Record the SD of $\bar{x}$ for $n = 25$ and for $n = 100$. What is the ratio? Why?

4. Bimodal population, $n = 50$: draw one sample. Is the sample normal? What is approximately normal?

3 Record use the tool and fill in the table

Sampling distributions

Draw 1,000 or more samples for each row. Simulated values vary a little; the key gives theory.

Population	n	Theory: SD of the statistic	Simulated SD	Shape
Skewed ($\sigma = 5.66$)	4			
Skewed ($\sigma = 5.66$)	36			
Uniform ($\sigma = 5.68$)	4			
$\hat{p}$ with $p = 0.1$	20			
$\hat{p}$ with $p = 0.1$	100			

4 Explain what did you discover?

1. The mean of the sampling distribution of $\bar{x}$ is _____, so $\bar{x}$ is an _____ estimator of μ.
2. SD($\bar{x}$) = _____. To cut it in half, multiply n by _____.
3. Central Limit Theorem: for large n, the sampling distribution of $\bar{x}$ is approximately _____ whatever the population's shape.
4. For $\hat{p}$, check Large Counts: _____ ≥ 10 and _____ ≥ 10.
5. A student says, "Because n = 50 is large, my sample is approximately normal." What is wrong with this statement?

5 Apply a new situation, no tool

1. Adult heights have μ = 66 in and σ = 4 in. For SRSs of size 16, find the mean and SD of $\bar{x}$.
2. Assuming heights are normal, find $P(\bar{x} > 67)$ for samples of size 16.
3. A skewed population has σ = 12. What sample size makes SD($\bar{x}$) = 2?
4. 8% of chips are defective. For samples of 150 chips, find the mean and SD of $\hat{p}$. Is a normal model reasonable?